JOSEPH GRUBER

Space Systems Engineering Manager | Technical Lead

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U.S. Citizen

Secret (<2 years inactive)

SUMMARY

Passionate, mission-driven, engineering leader with over 15 years of system engineering, engineering management, and project leadership experience in satellite ground systems, flight software, and spacecraft mission operations. I excel at leading multi-disciplinary teams in combining modern system & software engineering practices with a "agile aerospace" mindset and a strong bias for action to solve the engineering challenges of today's space industry.

EXPERIENCE

Blue Origin – Kent, WA — Feb 2019 – Present

Technical Project Manager

New Shepard Production Test & Operations

- Managed dedicated team of 10+ test software engineers developing test software to validate and qualify operational flight hardware.
- Developed and reviewed multiple test scripts for qualification and acceptance testing in coordination with test engineering team.
- Drove improvements to software development processes in alignment with mission and safety critical engineering standards.

New Shepard Software Architecture

- Guided 25+ software, network, GNC, and system integration engineers through design, development, and verification/validation of launch vehicle flight software.
- Coached team leads in building product roadmaps and managing backlogs resulting in first on-schedule flight software mission deliverable.
- Coordinated creation of multiple COTS software compliance plans documenting safety and reliability for software qualification review.
- Created automated collection and reporting of scope metrics and requirements verification via DOORS, improving communication and transparency to executive leadership and founder.
- Drove implementation with operations & maintenance team for continuous testing and deployment of flight software and integrated ground system.
- Documented and instituted workflows to manage scope for multiple payload and test missions coordinating parallel software release schedules.

Tech Stack: C++, Python, MATLAB, Jenkins, Gitlab, AWS, Ubuntu

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a.i. solutions – Lanham, MD — Jul 2016 – Feb 2019

Consulting Systems Engineer – Space Network SGSS

- Analyzed system architecture design as technical lead for upgraded TDRSS ground terminals and advised on engineering best practices for modernization and future sustainment operations of the space network operations center.
- Formulated lifecycle supportability risks and associated mitigation strategies from contractor system design and architecture documentation in preparation for operational handover post-final acceptance review.
- Created a configuration control system for long-term asset planning reporting for several thousand CSCI's.

Tech Stack: Oracle Database, WebLogic, & Application Server, VMware

EDUCATION

Embry-Riddle Aeronautical Univ.
Masters of Aeronautical Science / Space Studies

Florida Institute of Technology
B.A. Business Administration / Computer Information Systems

CERTIFICATIONS

AWS Certified Cloud Practitioner

AGI Satellite Tool Kit (STK) Master

Project Management Professional (PMP)

Professional Scrum Master (PSM-I)

AWARDS

NASA Flight Dynamics Support Services Team Technical Innovation Award

NASA/NOAA GOES-R Process Improvement and Innovation Award

Mission Systems Engineer – NASA Earth Science Mission Operations (ESMO)

- Managed, mentored, and developed as functional lead a team of 20+ multi-disciplinary system and software engineers supporting GNC of multiple NASA satellites and satellite situational awareness (SSA) of international earth science satellite constellation.
- Developed and executed $2.5 million project management plan with a consistent annual ~10% revenue growth.
- Designed and deployed CI/CD pipelines for flight dynamics software and constellation coordination software resulting in an 80% reduction of manual testing.
- Planned and implemented automated close approach violation and debris avoidance maneuver capabilities through automated acquisition of ephemeris data allowing for a 5x increase in daily conjunction assessments.
- Spearheaded process improvements and implemented corrective actions achieving CMMI-DEV Level 3 maturity.
- Created migration plan and led initiative to upgrade and modernize legacy, end-of-life software components ensuring reduced risk during increased lifespan of spacecraft.

Tech Stack: Node.js, C++, C#, Python, TeamCity, Bitbucket, FreeFlyer, VMware, Hyper-V, MATLAB

Hawaii Space Exploration Analog & Simulation (HI-SEAS) April 2013 – June 2018

Mission Support Lead (Research/Unpaid)

- Recruited and guided 30+ mission control team members on long-duration, isolated human spaceflight simulations as the primary communications and operational support interface to crew members.
- Developed frontend and backend services on AWS using Python and PHP for proto-flight distributed monitoring of habitat systems.
- Architected unique time-delayed communication protocols using C++ to simulate deep space digital communication.

ECG, Inc. – Greenbelt, MD Mar 2014 – Jan 2016

Lead Planner – NASA/NOAA Geostationary Operational Environmental Satellite R-Series (GOES-R)

- Directed team of 5 on the design, integration, and technical oversight of spacecraft, instrument, and ground segment schedules and statistical risk analysis of program scope.
- Contributed to architecture and development of petabyte-scale data storage solution for CCSDS-formatted level zero raw products used for spacecraft instrument calibration and validation.
- Generated plans and timelines for launch and orbit raising activities to validate on-orbit performance requirements and operational readiness of mission data products.
- Supported mission, flight, and launch readiness reviews on behalf of the chief engineer's office, creating and organizing schedule analysis and performance analysis reports.

Wyle Laboratories – Arlington, VA Feb 2012 – Mar 2014

Lead Planner – F-35 Joint Strike Fighter (JSF)

- Led four-person planning team for initial operational test & evaluation phase providing mentoring and coaching throughout integrated baseline reviews and system design & development flight testing.
- Conducted probabilistic risk analysis using Monte Carlo simulations to quantify technical and programmatic risks.
- Developed custom Python solutions for automated data analysis and metrics reporting.
- Witnessed and supported test events, simulations, and validation exercises, reporting results and metrics.

CSC – Washington, D.C. Feb 2011 – Feb 2012

Technical Project Manager – Department of Defense

- Led product development, deployment, and operations as engineering manager for a joint-agency communications/data application within a matrixed, multiple contractor organization.
- Reduced a six-month baseline slip to an on-time operational deployment for twelve remote sites by improving communications between engineers, regional technical staff, and program leadership.
- Managed five software & system engineers providing technical leadership throughout the full engineering lifecycle of requirements definition, configuration management, and verification & validation.

Early Career

- Systems Integration Engineer, SAIC & CTSC
- Systems Engineer, Lockheed Martin
- Senior System Administrator, RWBHD&M

PROFESSIONAL AFFILIATIONS

- Consultative Committee for Space Data Systems (CCSDS): Associate
- PMI; PMBOK 5th Edition: Risk Management Content Editor
- American Institute of Aeronautics and Astronautics (AIAA): SciTech Conference Chair
- AIAA National Capital Section (NCS): Communications Committee
- International Council on Systems Engineering (INCOSE)
- Toastmasters: President, Advanced Communicator, Advanced Leader

RELEVANT SKILLS

Platforms

Amazon Web Services (Compute, Database, Networking, Storage), Windows Server, Linux (Ubuntu, Red Hat), RTLinux, Mac OS X, Solaris, VMware, NASA core Flight System (cFS)

Software

a.i. solutions FreeFlyer, AGI Satellite Toolkit (STK), Atlassian (JIRA, Confluence, Bitbucket), GitHub, Gitlab, IBM Rational DOORS, IBM Rational Team Concert (RTC), Jenkins, Microsoft Project, Microsoft SQL Server, MySQL, MongoDB, MathWorks MATLAB, MathWorks Simulink, Oracle Primavera P6, Oracle Primavera Risk Analysis (PRA), OS/COMET, TeamCity

Programming Languages

Python, JavaScript, Node.JS, PHP, C#, C++, SQL, Visual Basic.Net, Visual Basic for Applications (VBA), Bash

Process

Agile methodology (Scrum, Kanban), Schedule Management, Test Driven Development, Automated Testing, Requirements Development, Release Planning, NPR 7150.2, NPR 7120.5, DO-178C, CMMI-DEV (Level 3)